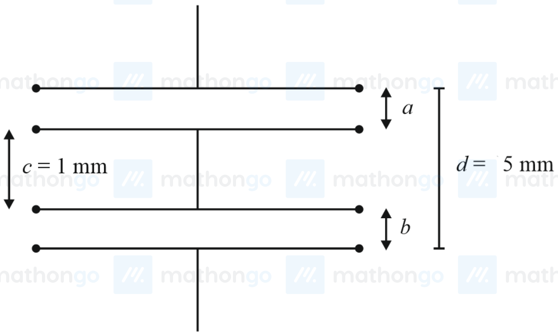
1. A parallel plate capacitor with plate area A and plate separation d is filled with a dielectric material of dielectric constant $K=4$. The thickness of the dielectric material is x, where $x<d$.

Let C_1 and C_2 be the capacitance of the system for $x=\frac{1}{3}d$ and $x=\frac{2d}{3}$, respectively. If $C_1=2\ \mu\mathrm{F}$, the value of C_2 is _____ μF.
[2023 (06 Apr Shift 1)]

2. As shown in the figure, two parallel plate capacitors having equal plate area of $200\ \mathrm{cm}^2$ are joined in such a way that $a \neq b$. The equivalent capacitance of the combination is $x\varepsilon_0 F$. The value of x is _____.



[2023 (06 Apr Shift 2)]

3. A $600\ \mathrm{pF}$ capacitor is charged by 200 V supply. It is then disconnected from the supply and is connected to another uncharged $600\ \mathrm{pF}$ capacitor. Electrostatic energy lost in the process is _____ μJ.
[2023 (08 Apr Shift 2)]

4. The equivalent capacitance of the combination shown is

[2023 (10 Apr Shift 1)]
(1) 2 C
(2) $\frac{5}{3}$ C
(3) $\frac{C}{2}$
(4) 4 C

5. The distance between two plates of a capacitor is d and its capacitance is C_1, when air is the medium between the plates. If a metal sheet of thickness $\frac{2d}{3}$ and of the same area as plate is introduced between the plates, the capacitance of the capacitor becomes C_2 . The ratio $\frac{C_2}{C_1}$ is
[2023 (10 Apr Shift 2)]
(1) $3\ :\ 1$
(2) $2\ :\ 1$
(3) $4\ :\ 1$
(4) $1\ :\ 1$

6. The electric field in an electromagnetic wave is given as
$\vec{E}=20\sin\omega\left(t-\frac{x}{c}\right)\vec{j}\ \mathrm{N\,C^{-1}}$, where ω and c are angular frequency and velocity of electromagnetic wave respectively. The energy contained in a volume of $5\times10^{-4}\ \mathrm{m}^3$ will be
(Given $\varepsilon_0=8.85\times10^{-12}\mathrm{C^2\,N^{-1}\,m^{-2}}$)
[2023 (11 Apr Shift 1)]
(1) 88.5×10^{-13} J
(2) 17.7×10^{-13} J
(3) 28.5×10^{-13} J
(4) 8.85×10^{-13} J

7. A parallel plate capacitor of capacitance 2 F is charged to a potential V. The energy stored in the capacitor is E_1. The capacitor is now connected to another uncharged identical capacitor in parallel combination. The energy stored in the combination is E_2. The ratio $\frac{E_2}{E_1}$ is

[2023 (11 Apr Shift 1)]

(1) $2 : 1$
(2) $2 : 3$
(3) $1 : 2$
(4) $1 : 4$

8. A capacitor of capacitance C is charged to a potential V. The flux of the electric field through a closed surface enclosing the positive plate of the capacitor is:

[2023 (11 Apr Shift 2)]

(1) $\frac{\mathrm{CV}}{\varepsilon_0}$
(2) Zero
(3) $\frac{2\,\mathrm{CV}}{\varepsilon_0}$
(4) $\frac{\mathrm{CV}}{2\varepsilon_0}$

9. In the given circuit. $\mathrm{C}_1=2\mu\mathrm{F}, \mathrm{C}_2=0.2\mu\mathrm{F}, \mathrm{C}_3=2\mu\mathrm{F}, \mathrm{C}_4=4\mu\mathrm{F}, \mathrm{C}_5=2\mu\mathrm{F}, \mathrm{C}_6=2\mu\mathrm{F}$. The charge stored on capacitor C_4 is ______ $\mu\mathrm{C}$.

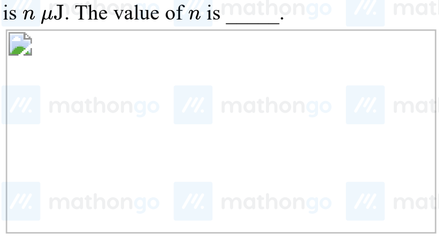
[2023 (11 Apr Shift 2)]

10. In the network shown below, the charge accumulated in the capacitor in steady state will be:

[2023 (13 Apr Shift 2)]

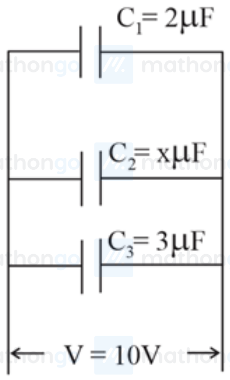
(1) $10.3\ \mu\mathrm{C}$
(2) $4.8\ \mu\mathrm{C}$
(3) $12\ \mu\mathrm{C}$
(4) $7.2\mu\mathrm{C}$

11. In the circuit shown, the energy stored in the capacitor is $n\ \mu\mathrm{J}$. The value of n is _____.



[2023 (13 Apr Shift 2)]

12. In the given figure the total charge stored in the combination of capacitors is 100 μC. The value of 'x' is ______________.



[2023 (15 Apr Shift 1)]

ANSWER KEYS

1. (3)	**2.** (5)	**3.** (6)	**4.** (1)	**5.** (1)	**6.** (4)	**7.** (3)	**8.** (1)
9. (4)	**10.** (4)	**11.** (75)	**12.** (5)				

1. (3)

The formula to calculate the capacitance of a parallel plate capacitor having thickness d and with a dielectric of thickness x between the plates is given by

$C = \frac{\varepsilon_0 A}{(d-x)+\frac{x}{K}} \quad \ldots (1)$

Substitute the values of the parameters for first case into equation (1) to obtain the capacitance.

$C_1 = \frac{\varepsilon_0 A}{\frac{2d}{3}+\frac{\frac{d}{3}}{4}}$

$= \frac{4\varepsilon_0 A}{3d} \quad \ldots (2)$

Similarly, substitute the parameters into equation (1) for the second case to obtain the capacitance

$C_2 = \frac{\varepsilon_0 A}{\frac{d}{3}+\frac{\frac{2d}{3}}{4}}$

$= \frac{4\varepsilon_0 A}{2d}$

$= \frac{2\varepsilon_0 A}{d} \quad \ldots (3)$

Divide equation (3) by equation (2) to obtain the capacitance in the second case.

$\frac{C_2}{C_1} = \frac{\frac{2\varepsilon_0 A}{d}}{\frac{4\varepsilon_0 A}{3d}}$

$= \frac{3}{2}$

$C_2 = \frac{3}{2} C_1 \quad \ldots (4)$

Substitute the value of the capacitance in the first case into equation (4) to obtain the value of the capacitance in the second case.

$C_2 = \frac{3}{2} \times 2\ \mu\text{F}$

$= 3\ \mu\text{F}$

2. (5)

The given data is

$A = 200 \times 10^{-4}\ \text{m}^2$

$d = 5\ \times 10^{-3}\ \text{m}$

The value of $a + b = 4 \times 10^{-3}$ m

The formula for a parallel plate capacitor is

$C = \frac{\varepsilon_0 A}{D}$

Let the capacitor with gap a be C_1 and capacitor with gap b be C_2

$C_1 = \frac{\varepsilon_0 A}{a},\ C_2 = \frac{\varepsilon_0 A}{b}$

The equivalent capacitance is

$\frac{1}{C_{eq}} = \frac{a}{\varepsilon_0 A} + \frac{b}{\varepsilon_0 A}$

$\Rightarrow C_{eq} = \frac{\varepsilon_0 A}{a+b}$

It is given that $C_{eq} = x\varepsilon_0$ F

So,

$\frac{\varepsilon_0 A}{a+b} = x\varepsilon_0$

$\Rightarrow x = \frac{A}{a+b}$

$\Rightarrow x = \frac{200\times10^{-4}}{4\times10^{-3}} = 5$

3. (6)

The formula to calculate the initial energy stored in the first capacitor is given by

$U_i = \frac{1}{2} C V^2 \quad \ldots(1)$

The formula to calculate the final energy stored when two capacitors are connected is given by

$$U_f = \frac{1}{2} C\left(\frac{V}{2}\right)^2 \times 2$$
$$= \frac{1}{4} C V^2 \quad \ldots(2)$$

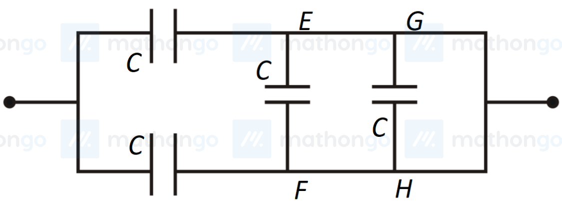
Subtract equation (2) from equation (1) to obtain the energy loss (ΔU).

$$\Delta U = \frac{1}{2} C V^2 - \frac{1}{4} C V^2$$
$$= \frac{C V^2}{4} \quad \ldots(3)$$

Substitute the values of the known parameters into equation (3) to calculate the required loss.

$$\Delta U = \frac{600 \times 10^{-12} \times (200)^2}{4} \text{ J}$$
$$= 6\ \mu\text{J}$$

4. (1)

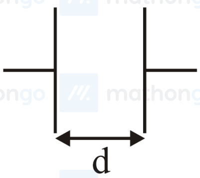


The potential difference across the branches EF & GH is zero. So, the equivalent capacitance will be only due to the combination of two capacitors in parallel.

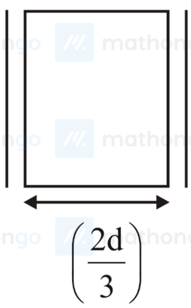
$C_{\text{eq}} = C + C$

$\Rightarrow C_{eq} = 2\text{ C}$

5. (1)



The formula of capacitance is $C_1 = \left(\frac{\varepsilon_0 A}{d}\right)$



The dielectric constant of metal sheet is $K_{metal} = \infty$

The new value of capacitance is

$$C_2 = \frac{\varepsilon_0 A}{d - t + \frac{t}{K_{metal}}} = \frac{\varepsilon_0 A}{d - \frac{2d}{3} + 0} = 3C_1$$

Clearly, the ratio,

$$\frac{C_2}{C_1} = \frac{3C_1}{C_1} = 3:1$$

6. (4)

Given that, $\vec{E} = 20 \sin \omega\left(t - \frac{x}{c}\right) \vec{\mathrm{j}}\ \mathrm{N\ C^{-1}}$

The average energy density of an electromagnetic wave is $\frac{1}{2}\varepsilon_0 E^2{}_0$.

The total energy stored is

$E_{\text{total}} = \frac{1}{2}\varepsilon_0 E_0^2 \times\ volume$

$\Rightarrow E_{total} = \frac{1}{2} \times 8.85 \times 10^{-12} \times \left(20\right)^2 \times 5 \times 10^{-4}$

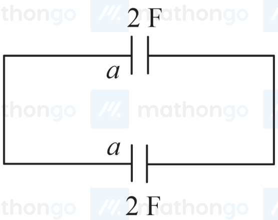
$= 8.85 \times 10^{-13}\ \mathrm{J}$

7. (3)

The charge on the plates of the first capacitor when connected against the potential difference V is given by

$Q = CV$

$= 2V$



When both the capacitors are connected, from the conservation of charge, it can be written that

$2V = 2V' + 2V'$

$\Rightarrow V' = \left(\frac{1}{2}V\right)$

where, V' is the new potential difference across each capacitor.

The formula to calculate the energy stored in the first capacitor is given by

$E_1 = \frac{1}{2} \times C \times V^2$

$= \frac{1}{2} \times 2 \times V^2$

$= V^2 \quad \ldots(1)$

For the second case, the energy stored in the combination of capacitor is given by

$E_2 = \frac{1}{2} nCV'^2$

$= \frac{1}{2} \times 2 \times \frac{V^2}{4} \times 2$

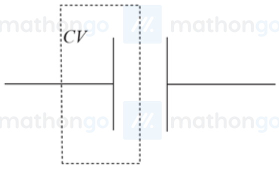
$= \left(\frac{V^2}{2}\right) \quad \ldots(2)$

Divide equation (2) by equation (1) to obtain the required ratio of the stored energy.

$\frac{E_2}{E_1} = \frac{\frac{V^2}{2}}{V^2}$

$= \left(\frac{1}{2}\right)$

8. (1)



The charge enclosed is given by $q_{inc} = CV$

Gauss' law states that the flux through any closed surface is equal to the charge enclosed by the surface.

Using Gauss' law the flux is, $\phi = \frac{q_{\text{inc}}}{\varepsilon_0} = \left(\frac{CV}{\varepsilon_0}\right)$

9. (4)

From the diagram of the circuit, $C_3,\ C_4,\ C_5$ are in series, so the total capacitance is

$\frac{1}{C'}=\frac{1}{2}+\frac{1}{4}+\frac{1}{2}$

$\Rightarrow C'=\frac{4}{5}\ \mu\text{F}$

C' & C_2 are parallel, the total capacitance is

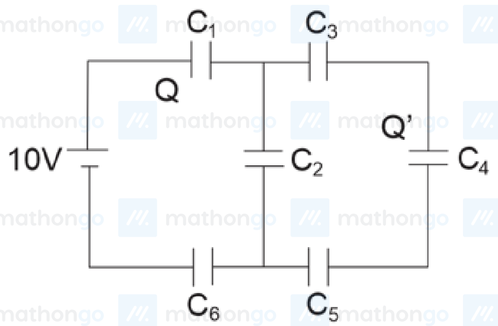
$C''=\left(\frac{4}{5}+0.2\right)\ \mu\text{F}=1\ \mu\text{F}$

$C_1,\ C'',\ C_6$ are in series. So the equivalent capacitance is,

$\frac{1}{C_{eq}}=\frac{1}{2}+1+\frac{1}{2}$

$\Rightarrow C_{eq}=0.5\ \mu\text{F}$

Using $Q=C_{eq}V$.



The charge through the battery, $Q=10\times\frac{1}{2}=5\ \mu\text{C}$

From the diagram above,

$Q'=\frac{5\mu\text{C}\times0.8}{0.8+0.2}=4\ \mu\text{C}$

10. (4)

In steady state, capacitor behaves as open circuit, so no current will flow.

The total resistance of the circuit is, $R=(6+4)\ \Omega=10\ \Omega$

The current flowing through the circuit is

$I=\frac{3\ \text{V}}{10\ \Omega}$

Potential difference on $6\ \Omega$ resistor

$V'=\frac{3}{10}\times6=1.8\ \text{V}$

The charge on the capacitor is,

$Q=CV'=4\ \mu\text{F}\times1.8\ \text{V}=7.2\ \mu\text{C}$

11. (75)

The capacitor works as an open circuit in the given configuration.

The potential difference between the two plates of the capacitor will be the difference across the resistors having resistances $3\ \Omega$ and $4\ \Omega$.

Hence, the potential difference (ΔV) across the capacitor can be calculated as follows:

$\Delta V=\left(\frac{4}{6}\times12-\frac{3}{12}\times12\right)\ \text{V}$

$=5\ \text{V}$

The formula to calculate the energy stored in the capacitor is given by

$U=\frac{1}{2}C\left(\Delta V\right)^2\quad\ldots(1)$

Substitute the values of the known parameters into equation (1) to calculate the required energy stored.

$U=\frac{1}{2}\times6\ \mu\text{F}\times(5\ \text{V})^2$

$=75\ \mu\text{J}$

12. (5)

All the capacitors are in parallel combination and therefore potential drop across each capacitor would be the same.
Charge on the capacitor can be written as, $Q = CV$.
The charge on C_1 is $Q_1 = 2 \times 10 = 20\ \mu\text{C}$
The charge on C_2 is $Q_2 = x \times 10 = 10x\ \mu\text{C}$
The charge on C_3 is $Q_3 = 3 \times 10 = 30\ \mu\text{C}$
By the given data.
$C_1 + C_2 + C_3 = 100\ \mu\text{C}$
$\Rightarrow 10(2 + x + 3) = 100$
$\Rightarrow x = 5$